Library Book Management System

Project Report

1. Project Overview

The Library Book Management System is an application that helps people who work at libraries manage their books. This system lets staff add books keep track of how many copies are available lend and return books and look at all the books in the library. You can do all of this using an interface on your computer. You do not need to be connected to the internet to use this system.

2. Objectives

- We want to make a tool that libraries can use to keep track of their books.
- We want to make sure that the parts of the system that do things are separate.
- We want to make a core part of the system that can be used in systems.
- We want to make sure the system does not stop working when someone types in something

3. Project Structure

The project has two parts:

```
library/
    book_manager.py   # This is where all the library logic is
    main.py           # This is where you start the application
    README.md         # This is where we keep information about the project
```

The book_manager.py file has all the code that makes the library system work. It has four functions: add_book, borrow_book, return_book and list_books.

The main.py file is where the application starts. It talks to the user does what they ask and then shows them the results. It also helps when the user types in something

4. Features

Feature	Description
Add Books	You can add a book or add more copies of a book that is already in the system.
Borrow Books	You can lend a book if there is one
Return Books	You can return a book that you borrowed.
List Books	You can look at all the books in the library.
Error Handling	The system also handles mistakes.

5. Data Model

If you type in something that's not a number the system will ask you to try again.

The Library Book Management System stores books in a dictionary. Each book has the following information:

Field	Type	Description
author	str	This is the name of the person who wrote the book.
available	int	This is how many copies of the book are available to lend.
total	int	This is the total number of copies of the book, in the library.

6. Known Limitations

- The system does not save any information when you close it.
- The system does not keep track of who borrowed a book.
- The system is case-sensitive so it treats "book" and "Book" as books.
- The system is designed for one person to use at a time.

7. Future Improvements

- We want to save the library data in a file so it is not lost when the system closes.
- We want to add user accounts so people can log in and see what books they borrowed.
- We want to add a search function so people can find books by author or keyword.

- We want to add dates and reminders so people know when to return a book.
- We want to make a web version of the system many people can use it at the same time.

8. How to Run

You need to have Python 3.6 or higher to run the system. You do not need any programs.

You can get the system by doing the following:

```
git clone https://github.com/your-username/library-book-manager.git
cd library-book-manager
python main.py
```

When you run the system you will see a menu. You can type a command and press Enter. The system will guide you through each action. Ask you for more information if needed.